Applying Model-Driven Development to Business Systems using RM-ODP and EDOC

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Abstract

Improving development efficiency and maintainability for business systems requires seamless development process, and both RM-ODP and MDA play a key role to this end. This paper shows our Model-Driven Development process in building business systems using RM-ODP and UML Profile for EDOC, with a case study of Electronic Health Record system models, and discusses several issues related to RM-ODP standard.

1. Introduction

As business systems are getting complex and large in recent years, their development efficiency and maintainability need to be improved. Especially, efforts to Total Optimization by SCM (Supply Chain Management) have a great influence on the way business systems are developed. To conduct Total Optimization, various information systems introduced to enterprise and supply chain need to collaborate with each other. In order to develop such information systems, it is required to model business processes and identify the roles of those systems in the entire business. Moreover, certain mechanism is necessary to transform the models seamlessly into implementation, which makes systems development more efficient with traceability among models.

2. MDA and RM-ODP Viewpoints

MDA, advocated by OMG (Object Management Group), is the technology that seamlessly reflects models to implementation. In MDA, models are developed from three perspectives: CIM (Computation Independent Model), PIM (Platform Independent Model), and PSM (Platform Specific Model). MDA is the abstract framework, thus concrete development processes are required for realization of MDA.

This paper shows overview of our Model-Driven Development process using RM-ODP [1] framework and notations defined by UML Profile for EDOC[2] (referring to it as EDOC from now on). Our process uses RM-ODP Viewpoints as follows: business models are developed in Enterprise Viewpoint, information models in Information Viewpoint, and component models in Computational Viewpoint. These models are CIMs and PIMs in MDA. System architectures are defined in Engineering Viewpoint, and mapping rules are defined in Technology Viewpoint.

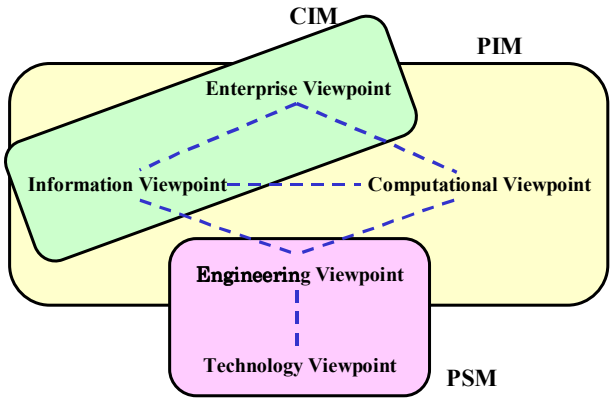

Figure 1. Relation between MDA and Viewpoints of RM-ODP

In the following sections, Electronic Health Record system models[3][4] are used as a case study for applying this development process. This case study is an accomplishment of a project formed by two organizations: JAHIS, whose focus is on establishing approaches for developing standard components for Electronic Health Record systems, and INTAP, whose focus is

[1] The standard frameworks for distributed object systems, standardized by ISO/IEC and ITU.
[2] The set of extended UML for distributed object systems, standardized by OMG It extends notations for components and business processes.
[3] The Enterprise Viewpoint part of these models are an accomplishment of "The Development of Electronic Health Record System through Standardizing Components" as a special science research theme awarded by the Ministry of Health, Labor and Welfare in the fiscal year 2002.
[4] The scale of those models includes 12 communities and 32 business processes.

on interoperability among enterprise systems with RM-ODP. Japanese government has been involved in and funds the project.

3. Enterprise Viewpoint

In Enterprise Viewpoint, target business models and system requirements are modeled. The most important artifacts of this Viewpoint are business processes. Organizing business processes in the entire supply chain and enterprise clarifies the role of information systems. The first thing is to define a scope for the system and divide it into smaller ones for unit called Community. It is a unit to organize the scopes with their purposes. Defining Communities can make the size of the scope appropriate. Next, procedures for accomplishing purposes of each Community are defined as business processes. With Business Process Profile and Event Profile of EDOC, not only event-driven business process can be described, but responsible parties of activity as well as information and functions used in the business process can be defined as Process Role.

In this viewpoint, we did not make much use of developed policy statements. The issue was lack of policy statement language with clear syntax, semantics, and grammar. The emergence of policy statement language, and the structuring rule for policies with other enterprise model elements in the enterprise viewpoint specification are expected.

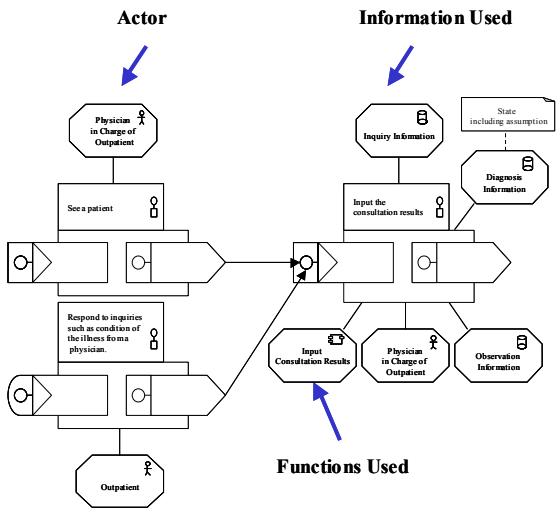

Figure 2. Example of Business Process (Part of Outpatient Consultation)

4. Information Viewpoint

In Information Viewpoint, information referred to and processed by the systems is modeled. Our process defines that the details of information used in the business processes should be analyzed. The resulting information model elements of the system are described using Entity Profile in the EDOC. The use of the profile enables to describe primary key and foreign key of entity in RDB, and determine access path to information to create information components (referring to it as entity components from now on). Constructing information as components is effective in preventing reduction of performance, since it identifies the gate of information to control the number of remote access.

In this viewpoint, we did not consider dynamic addition or deletion of model elements. However, if dynamic change is included in Enterprise Viewpoint Model, such as dynamic creation of a community with new roles introduced, information viewpoint model should also accommodate corresponding dynamic changes within the model..

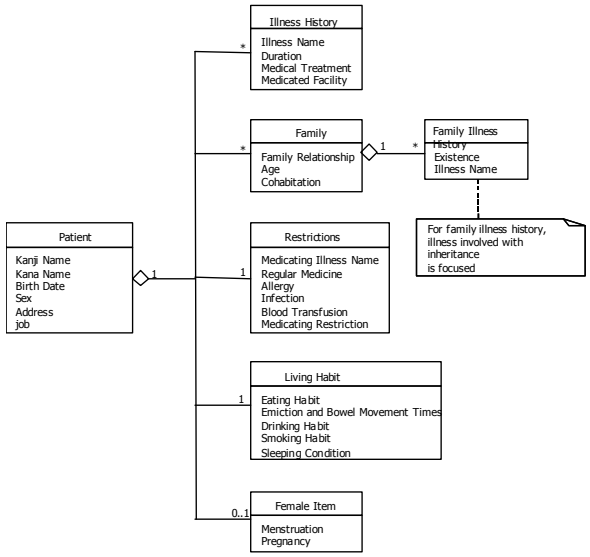

Figure 3. Example of Information Model (Patient)

5. Computational Viewpoint

In Computational Viewpoint, functions of systems are modeled. Our process defines that the details of functions used in the business processes (i.e. performers and artifacts) and entities should be analyzed. The functions of systems are described as process components of CCA (Component Collaboration Architecture) Profile in the EDOC. If collaboration exists among enterprises and/or among information systems, it is described as interaction among components. Note that interfaces of components are discovered based on analysis of Information Viewpoint.

In the EDOC, collaborations are described by connecting components through Ports. If the functions of systems require persistent information (database information), the process components are connected to the entity components. Operation procedure of the connected components (both process and

entity) is defined as Protocol, and described using State chart diagrams of UML.

It should be noted that in RM-ODP the basic unit for encapsulation is object, and the interaction between objects are made by signal, operation and flow. Although we believe CCA ProcessComponent can represent Computational Object in pragmatic way, it would be preferable to have necessary modeling concepts for components and interactions between them (messaging) in RM-ODP itself.

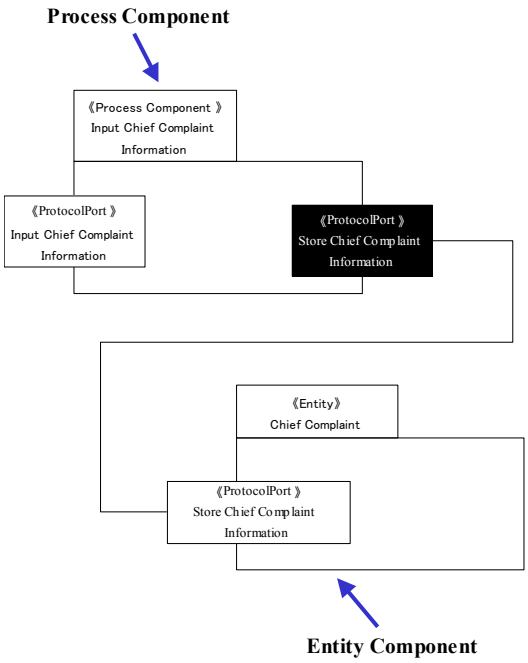

Figure 4. Example of Component Model

6. Engineering and Technology Viewpoint

In Engineering Viewpoint, system architectures are designed. The design includes deployment configuration of components, transaction, and security.

It should be noted that there are choices in mapping Component Model onto platforms. One choice is on platform styles, and another choice is on configuration of nodes. Figure 5 was our choice for pilot implementation, but there were other possibilities, e.g. client node could host only GUI application and network interface components, and all other components may be hosted on one or more server nodes. Also the same component model could have been mapped onto web services platform or CORBA platform.

Another observation is that it is this viewpoint that shows portions of human-system interaction with Client node box. The model of human-system interaction should also be described in relevant viewpoints in addition to the Engineering viewpoint model.

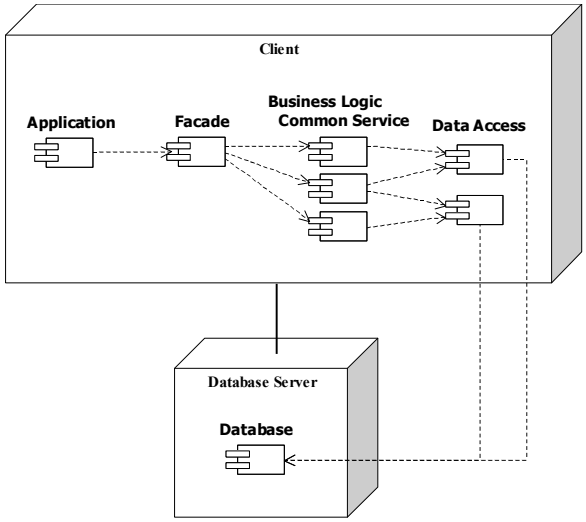

Figure 5. Example of Deployment Configuration of Components

In Technology Viewpoint, PIM as the artifact of all the processes is mapped to a specific platform to derive PSM from it. The following example is one of mapping rules of EDOC to .NET. The mapping rules were used in the pilot development of Electronic Health Record system.

EDOC (PIM)	.NET (PSM)
Process Component 《ProcessComponent》	Mapped to .NET assembly. Select assembly type according to requirement such as deployment, performance, and security.
Protocol 《Protocol》	Mapped to interface.
Interface 《Interface》	Mapped to interface.
Protocol Port 《ProtocolPort》	Mapped to interface.
Operation Port 《OperationPort》	Initiator: mapped to method call. Responder: mapped as interface method.
Flow Port 《FlowPort》	In case of flow port included in operation port: Initiator: mapped to argument of method. Responder: mapped to returned value of method.
Entity Component 《Entity》	Mapped to .NET component using ADO.NET.
Entity Data 《EntityData》	Mapped to ADO.NET DataSet defined by XML schema.
Key 《Key》	Mapped to constraint defined by XML schema.
Foreign Key 《ForeignKey》	Mapped to constraint defined by XML schema.

Table 1. Example of Mapping Rules (EDOC to .NET)

In Table 1, interfaces provided by the process components and various Ports included in the components are mapped to interfaces of .NET components and their methods. The entity components are mapped to .NET components that use functions of ADO .NET. Remote accesses among components may be required depending on deployment configuration of components in Engineering Viewpoint. In this case, Web services and remoting functions of ASP .NET should be used to realize the remote accesses.

7. Discussion

Based on the experience of above explained project, it is the authors' belief that the following issues are to be considered at the workshop.

7.1 Viewpoint correspondence, CCA, and Computational Language

In RM-ODP and Enterprise Language standard, viewpoint correspondences are defined but are not very useful in practice. In EDOC, because of its design, Business Process, Entity, and Event are defined as specializations of CCA, and thus there exists stronger correspondence between those and CCA. Once the mapping from those to CCA is done, and the final model is represented in CCA, the resulting model should be considered as Computational Viewpoint Model. However there are some issues.

1) Computational Language does not have concept of "component."

2) Computational Language does not have adequate concept for message-based communication between components.

If those issues were resolved (e.g. by adding those concepts or interpretations to RM-ODP), RM-ODP, EDOC, and probably UML2 too, would synchronize more tightly and improve the modeling capability.

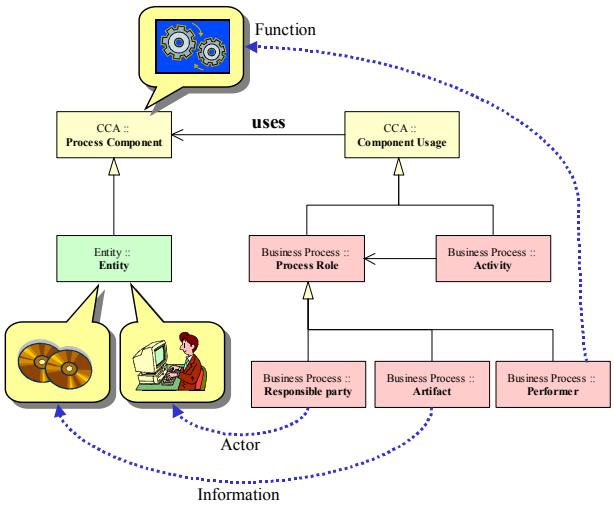

Figure 6. Relationship of Profiles

7.2 Dynamic evolutions of viewpoint models

In Enterprise Language, dynamism such as dynamic creation of a community is described. However, dynamism should also be considered in other areas and viewpoints, because we will need some means to describe "as-is to to-be" model evolutions or migration in various occasions, e.g. when designing new system model to include additional viewpoint (e.g. information) objects to the existing model.

7.3 Human-System interactions

Although one of the classes of reference points defined in RM-ODP Part 2 is perceptual reference point, there is no specific viewpoint concepts dedicated for this purpose. We believe it necessary to have text in RM-ODP suggesting which viewpoint this belongs to and how it should be described (concepts and structuring rules). One observation in EDOC is that it is likely to have Human-System interactions' binding to CCA component representing user interface.

7.4 Policy language

Although it may not be directly related to RM-ODP, as discussed in clause 3, we would like to see a language for policy statement definitions.

8. Summary

8.1 Model-Driven Development

Our concrete process worked well to develop business systems by Model-Driven Development, resulting in successfully applying our process to Electronic Health Record system. It turned out that applying development frameworks of RM-ODP enabled seamless development from business models to program codes. Especially in developing business systems that aim for Total Optimization, this framework is certainly the best since the scope of collaboration can be considered as distributed objects. The use of EDOC also enables to describe various aspects of business systems in detail. As EDOC has great affinity for RM-ODP, it is the most appropriate modeling language in building CIMs and PIMs. Since this case study of Electronic Health Record system is implemented as the pilot system, our next goals are to use our process to implement Electronic Health Record system of practical scale, and to apply our process to systems with other domains for its refinement.

8.2 Possible revisions to the standard

We have identified several issues for RM-ODP standard, and included some of our suggestions in this paper. It is our hope that those issues be considered, discussed, and resolved at the workshop, or passed to ISO/IEC and ITU-T RM-ODP group as a part of issues list for RM-ODP revision work.

Reference

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[2] MDA Guide, Document number: omg/03-06-01

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[4] UML Profile for EDOC Part I, Document number: ptc/2004-02-01

[5] UML Profile for EDOC Part II, Document Number: ad/01-08-20

[6] JAHIS Web Site, http://www.jahis.jp/english/english.html

[7] INTAP Web Site, http://www.net.intap.or.jp/e/